
Evaluating Signal Dilution and Model Efficiency with BiLSTM Models Trained on Whole Phage Proteomes

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Abstract

The mass consumption of antibiotics is required to treat bacterial infections and diseases, resulting in bacteria gradually evolving to exhibit resistance to conventional antibiotic treatments. This worldwide issue is referred to as antimicrobial resistance (AMR), with an estimated 4.7 million deaths. Bacteriophage therapy serves as a promising alternative to antibiotics, where viruses that infect bacteria (bacteriophages) are used to kill bacteria. Various machine learning attempts have been made to predict bacteriophage lifestyle. However, conventional tools rely on comparing known and conserved protein domains. In this study, we evaluate training a bidirectional long-short term memory (BiLSTM) recurrent neural network (RNN) architecture on whole proteomic sequences, achieving a 0.5045 Matthews Correlation Coefficient score, a 75% weighted F1 score, and a 75% accuracy rate. We also address the caveats that arise, notably with lifestyle marker gene signal dilution, and accuracy discrepancies that arise with whole-proteomic deep learning.

1 Introduction

Adopting current bacterial treatment approaches requires mass consumption of antibiotic treatments as a way of resolving bacterial infections. The collective overuse of antibiotics and other medications has led to various varieties of bacterial species that are no longer affected by antibiotics, because diverse groups of bacteria are gradually evolving to be resistant to antibiotic treatments. This global crisis is commonly referred to as antimicrobial resistance (AMR), with an estimated 4.7 million deaths associated with AMR [1]. An alternative towards traditional alternatives involves using bacteriophages (phages), viruses that infect certain bacterial strains. Bacteriophages are typically classified between virulent and temperate groups. Virulent bacteriophages utilize the virulent cycle, in which the bacteriophage lyses the bacterial host cell. Temperate phages either lyse the bacterial cell, or integrate into the bacterial host cell's genome for further replication [2]. Predicting phage lifestyles for such appropriate therapy is an intensively laborious method, which makes it often impractical to do so with new phages.

Prior classification tools have made substantial progress on classifying such lifestyles. Tools such as, PHACTS (Phage Classification Tool Set), analyze bacteriophage protein sequences with sequence similarity algorithms against a subset of characterized proteomes. Initially, PHACTS creates a set of randomly selected query proteins from the now defunct PHANTOME Database, i.e. randomly selected. Alternatively, BACPHLIP (BACTERiophage Lifestyle Predictor) uses HMMER3 to search the inputted bacteriophage genome for conserved protein domains, and feeds each bacteriophage's profile through a Random Forest model to chance a prediction. Training sets in such were constructed by producing a set of N similarity vectors, where a percent identity score is calculated between the individual bacteriophage's proteome and the random set of query protein sequences, utilizing the

FASTA35 program. The tool further uses a Random Forest classifier with decision trees, reaching a 99% precision rate [3].

However these models rely on identifying and comparing known, conserved protein domains across tested phages, leading both models to not account for bacteriophages that have uncharacterized or sequences that have diverged over time, leading such models to possibly misclassify bacteriophage lifestyles. Additionally, both BACPHLIP & PHACTS look for lifestyle marker genes such as integrases and repressors. If a bacteriophage with a fragmented or short contig sequence is tested, the missing sequence data may contribute towards a false, and potentially misclassified output.

Here, we present a hybrid CNN-BiLSTM neural network designed to capture local motif signatures, while also capturing sequential and long-range dependencies across each bacteriophage's proteome, serving as an adaptable alternative, which can infer lifestyle, as compared to domain-based Random Forest classifiers. The model was trained on whole bacteriophage proteomes, with a compiled dataset consisting of 172 temperate bacteriophages, and 151 virulent phages.

However, our model achieves a decently low weighted F1 score of 75.0%, while achieving a similar 75.0% accuracy, revealing several important distinctions about applying sequential-based deep learning architectures to 1-dimensional biological sequences, such as amino acids which make up a proteome. We uncovered this imbalance to be due to a main critical reason:

1. Whole bacteriophage proteomes consist of thousands of genes, which are each made up of hundreds of amino acids. When a few specific genes that heavily influence lifestyle, such as an integrase or repressor gene is processed alongside hundreds of other genes, deep learning architectures (e.g. LSTMs and RNNs) will struggle with long term sequence dilution.

This paper will evaluate the model made, and the important caveats and limitations that come with training deep learning architectures to predict binary bacteriophage lifestyle, on whole full-size proteomes.

2 Methods

2.1 Proteome Collection

For this study, we used the dataset from the BACPHLIP public Zenodo repository [5], using the "processed_benchmark_set.csv" CSV file, which contained 507 phages. From the file, each bacteriophage was listed with an accession number and an empirically proven lifestyle. To We compiled a Python script utilizing the Biopython Entrez tool [6], to fetch (using the accession no.) and append the proteome sequence (X variable) to a CSV column. After this, training, validation, and testing sets were created with a stratified 60 / 20 / 20 split into 3 datasets, train_strat.csv, test_strat.csv, and val_strat.csv, such that each dataset was to receive at minimum one phage of a specific genera (training receives 50%, validation & testing receives 25% each), with the same proportion. Each of the protein sequences (X variable) were imported through a Python dictionary, and passed through a text-vectorization layer. Binary lifestyles (Y variable) were subsequently stored as arrays.

2.2 Model Architecture

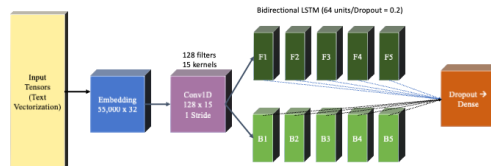


Figure 1: Displays the model architecture, and pipeline flow.

We implemented our model architecture using the Keras Sequential API (See Fig. 1). The network accepts a dynamic tensor shape, made by adapting a Text-vectorization layer to each proteome. Our architecture begins with an initial embedding layer consisting of a custom calculated input dim, and an output dim of 32. The next layer is a 1D-convolutional layer, with 128 filters and a kernel size of 15. The next layer is a BiLSTM, containing 64 hidden units and a hyperbolic tangent (tanh) main activation function, and a sigmoid recurrent activation function for the internal gates (forget, input, output) [7]. We then apply a dropout layer to prevent overfitting, with a rate of 0.6 (60%). We finally apply a single dense unit, with a sigmoid activation for binary classification.

3 Results

3.1 Classification Metrics

The model was evaluated using the classification-report, matthews_corrcoef, and PrecisionRecallDisplay API, made available through the scikit-learn library, providing model reliability and testing metrics necessary for analysis.

3.1.1 Classification Report

The classification report outputted is shown in Table 1. The Matthews Correlation Coefficient (MCC) Score resulted to be 0.5405 (54.05%)

Table 1: Classification Report

	Results			
	precision	recall	f1-score	support
Temperate	0.89	0.63	0.74	52
Virulent	0.65	0.90	0.76	40
accuracy			0.75	92
macro avg	0.77	0.77	0.75	92
weighted avg	0.79	0.75	0.75	92

3.1.2 Precision-Recall Curve Graph

The Precision-recall curve (PRC) was constructed from the model's prediction's Y testing binary labels, against the Y prediction probabilities (See Figure 2).

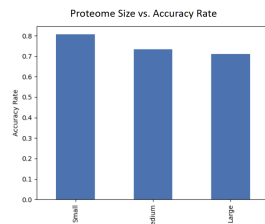


Figure 2: Proteome size vs. accuracy bar graph, first failure mode evaluation.

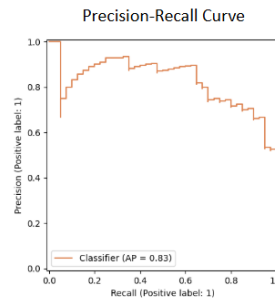


Figure 3: Precision-recall curve graph, second failure mode evaluation.

3.1.3 Proteome Size vs. Accuracy Rate Graph

The Proteome Size vs. Accuracy Rate Graph was constructed by dividing proteome groups into 3 quartiles, based on length, and calculating the mean accuracy for each (See Figure 3).

4 Analysis

Throughout this, our model struggled to ideally identify and classify temperate phages, being able to classify temperate phages reliably with an 89% precision, but missed a significant amount of missed cases, yielding a 63% precision. Conversely, the model yielded an excellent rate at catching virulent cases, with a 90% recall, but struggled with a high amount of false positives (65% precision).

Additionally, the model also recieved a Matthews Correlation Coefficient score of 54.05% (0.5405). On a scale from -1 to +1, with -1 indicating a perfect random guess, and +1 indicating a flawless prediction. Unlike other classification metrics, the MCC score does not reward highly imbalanced datasets, primarily by interpreting true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), using Equation 1, indicating that our model’s predictions and the testing data had a moderate positive correlation, performing better than a completely random guess.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (1)$$

Furthermore, we evaluated our model using a comparison of the size of the proteome against the accuracy rate of the model on each group. The size of the proteome was grouped into three bins containing an equal amount of phages, being divided at the 33rd and 66th percentiles of length. The resulting graph (Figure 2) indicates that as the proteome length of the group increases, the model prediction accuracy decreases; with smaller groups earning a $\sim 82\%$ accuracy, medium groups earning a $\sim 74\%$ accuracy, and large groups earning a $\sim 72\%$ accuracy.

Additionally, we also evaluated our model using a precision-recall curve (PRC), as shown in Figure 2. The PRC graph displays a jagged curve with rapid drops, while also mostly residing in the top right corner, demonstrating a well balance between recall and precision rates. Additionally, the classifier average precision (AP) rate is 83%, indicating a high predictive accuracy rate when predicting positive classes. These results indicate that the model achieves a decently high predictive power, especially when compared to a baseline AP of 0.50 and a perfect AP of 1.0.

However, through these results, our BiLSTM model does not yield dependably high metrics which would make this tool useful for phage therapy applications. Bacteriophage therapy requires extremely precise and reliable fine-picking when selecting a group of bacteriophages to use for clinical treatment. If an incorrect or inappropriate phage is predicted to be virulent, and used in therapy, it may pose severe hazardous risks on the patient. Additionally, even slightly flawed predictions can alter the reproducibility and credibility of scientific experiments

These results illustrate interesting correlations between specific groups of phage proteomes. As shown in Figure 2, accuracy decreases at the proteome size increases, reaching a low accuracy rate of $\sim 72\%$. This downward trend shows that BiLSTMs cannot efficiently be used to classify lifestyle on relatively longer sequences.

Moreover, longer sequences introduce extra proteomic information. In bacteriophages, a specific collection of marker genes that dictate lifestyle (e.g. integrases and repressors) are traditionally known to dictate bacteriophage lifestyle. With the processing of extra proteomic information alongside these genes that tend to mainly indicate lifestyle, these marker genes’s states get balanced out or diluted across the gradient, while the BiLSTM attempts to incorporate extra proteomic information. Instead, the model would incorporate abundant and commonly shared genes essential to bacteriophage function as a whole, reducing the marker gene’s signals.

Therefore, we call for further deep learning efforts to not attempt to process whole proteomeic information per phage, but instead use mechanisms where marker genes receive more contextual importance, such as attention layers [10], which can allow the model to weigh and heed importance to certain genes instead.

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5 LLM Usage Disclosure

During the preparation of this work, the author(s) utilized Google Search’s AI Mode and OpenAI’s ChatGPT to simply assist with debugging research code, resolving errors, help guide paper submissions, and write simple scripts. These tools were not used to fabricate or manipulate data, nor were they used to write the paper. After using these services, the author(s) reviewed the output as needed to take full responsibility for the integrity of the work.